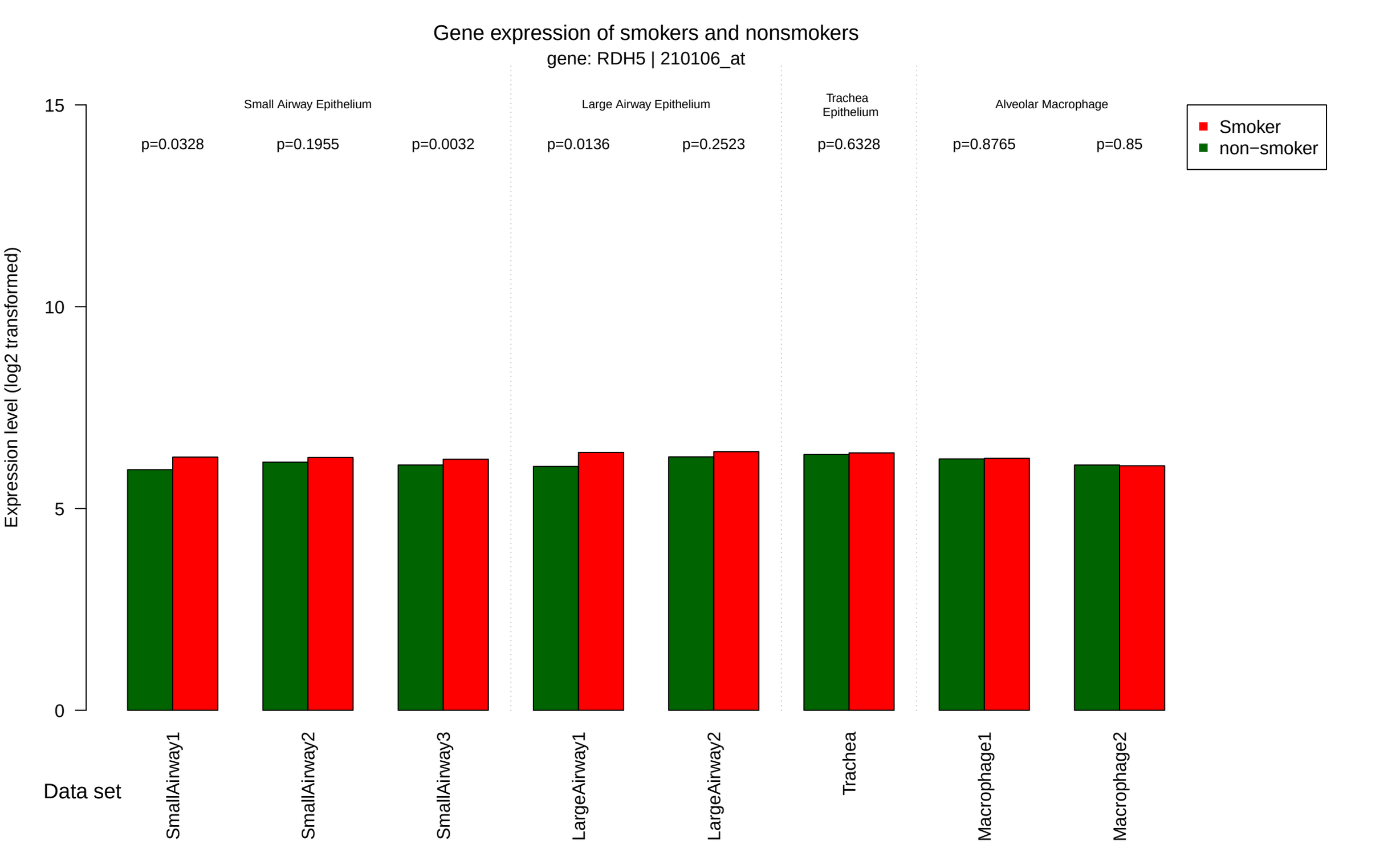
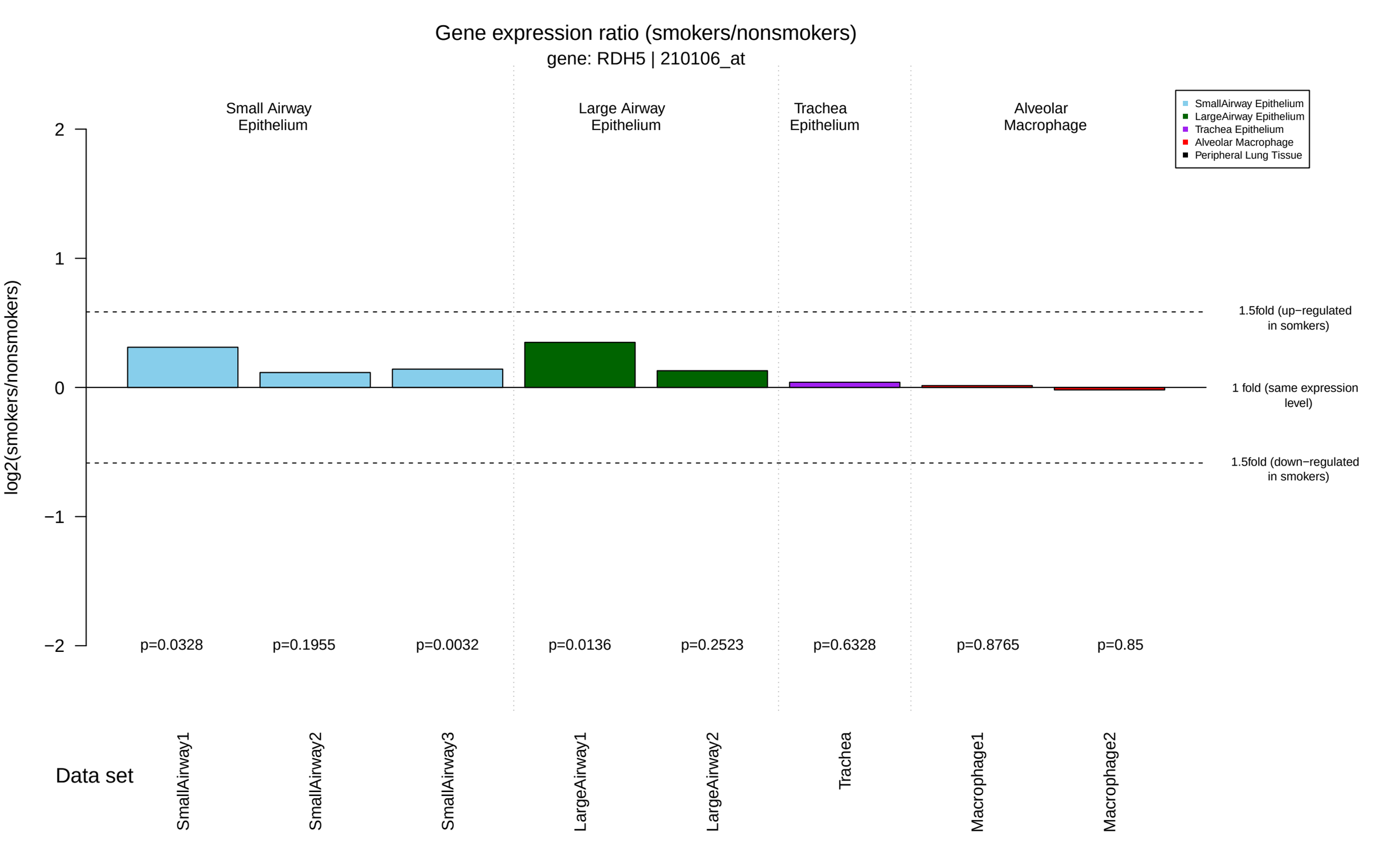
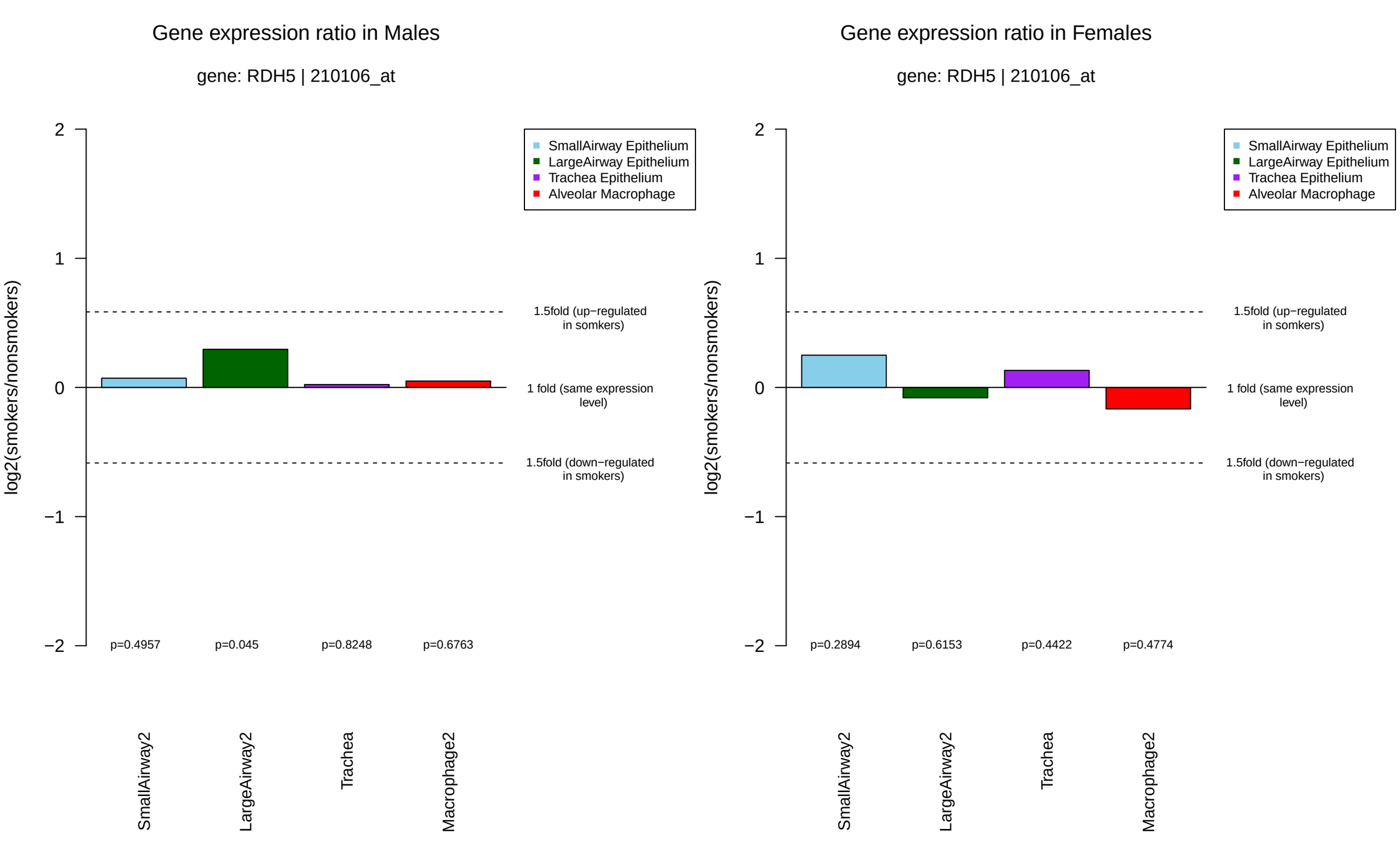
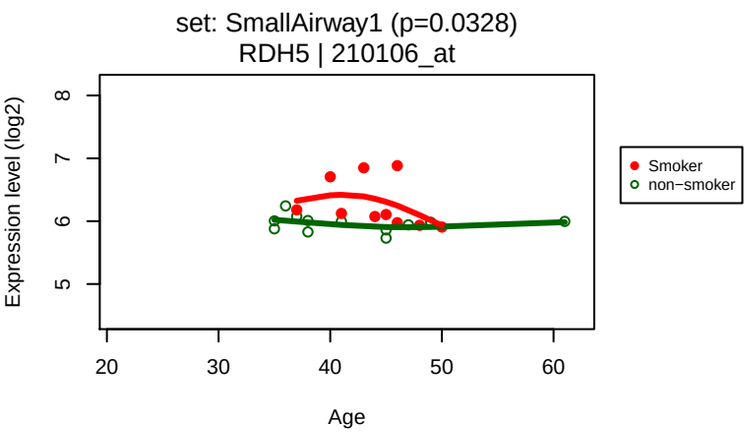
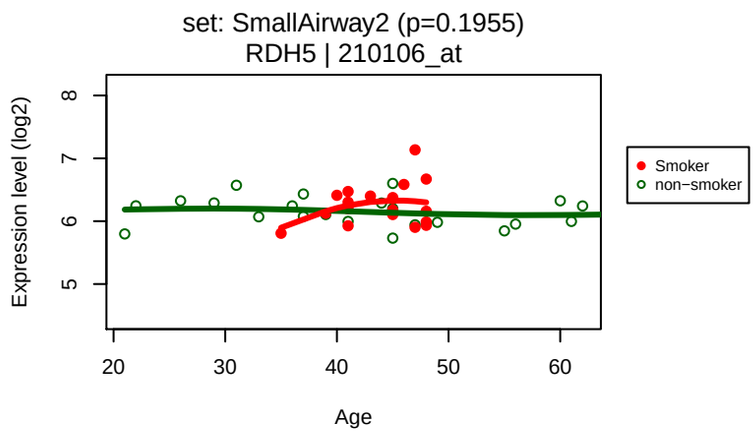
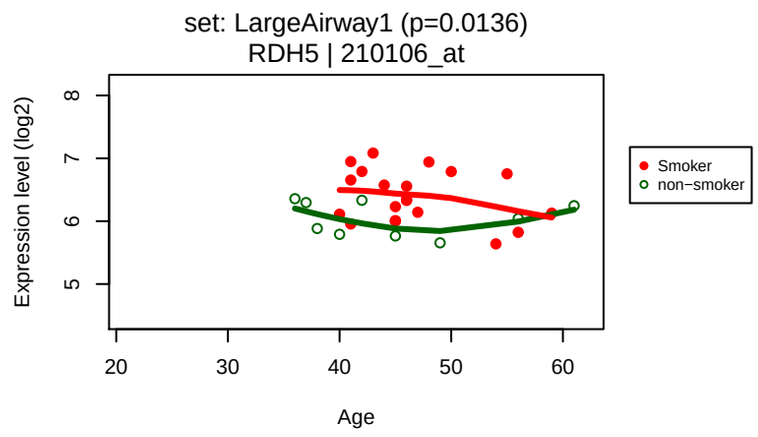

Gene expression of smokers and nonsmokers
gene: RDH5 | 210106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0328
p=0.1955
p=0.0032
p=0.0136
p=0.2523
p=0.6328
p=0.8765
p=0.85
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

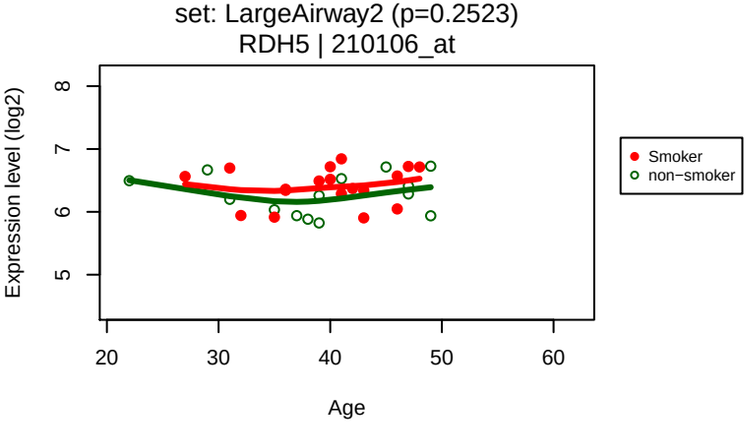

Gene expression ratio (smokers/nonsmokers)
gene: RDH5 | 210106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0328
p=0.1955
p=0.0032
p=0.0136
p=0.2523
p=0.6328
p=0.8765
p=0.85
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

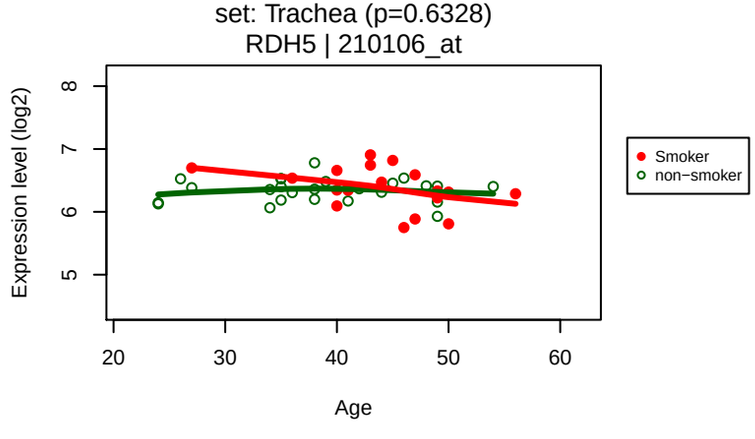

Gene expression ratio in Males
gene: RDH5 | 210106_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4957
p=0.045
p=0.8248
p=0.6763
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RDH5 | 210106_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2894
p=0.6153
p=0.4422
p=0.4774
SmallAirway2
LargeAirway2
Trachea
Macrophage2

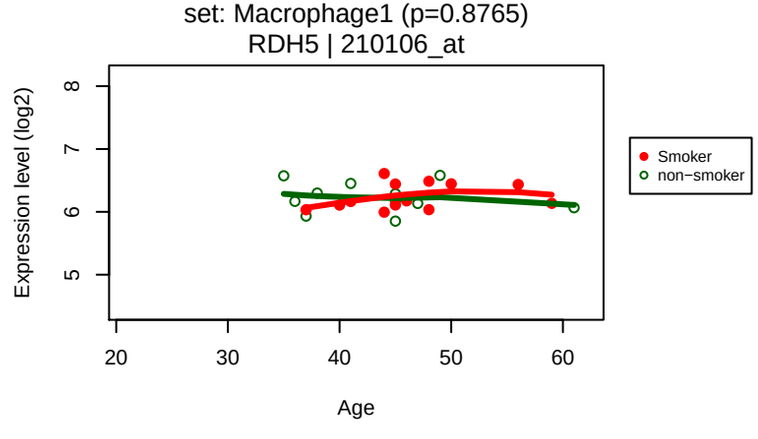

set: SmallAirway1 (p=0.0328)
RDH5 | 210106_at
Expression level (log2)
Age
Smoker
non−smoker

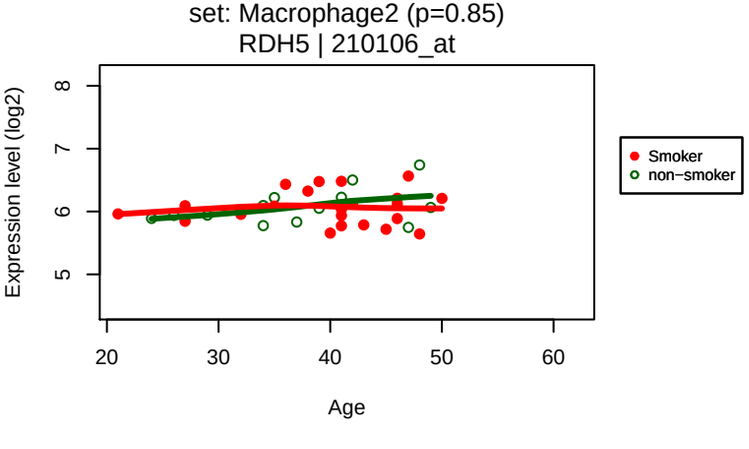

set: SmallAirway2 (p=0.1955)
RDH5 | 210106_at
Expression level (log2)
Age
Smoker
non−smoker

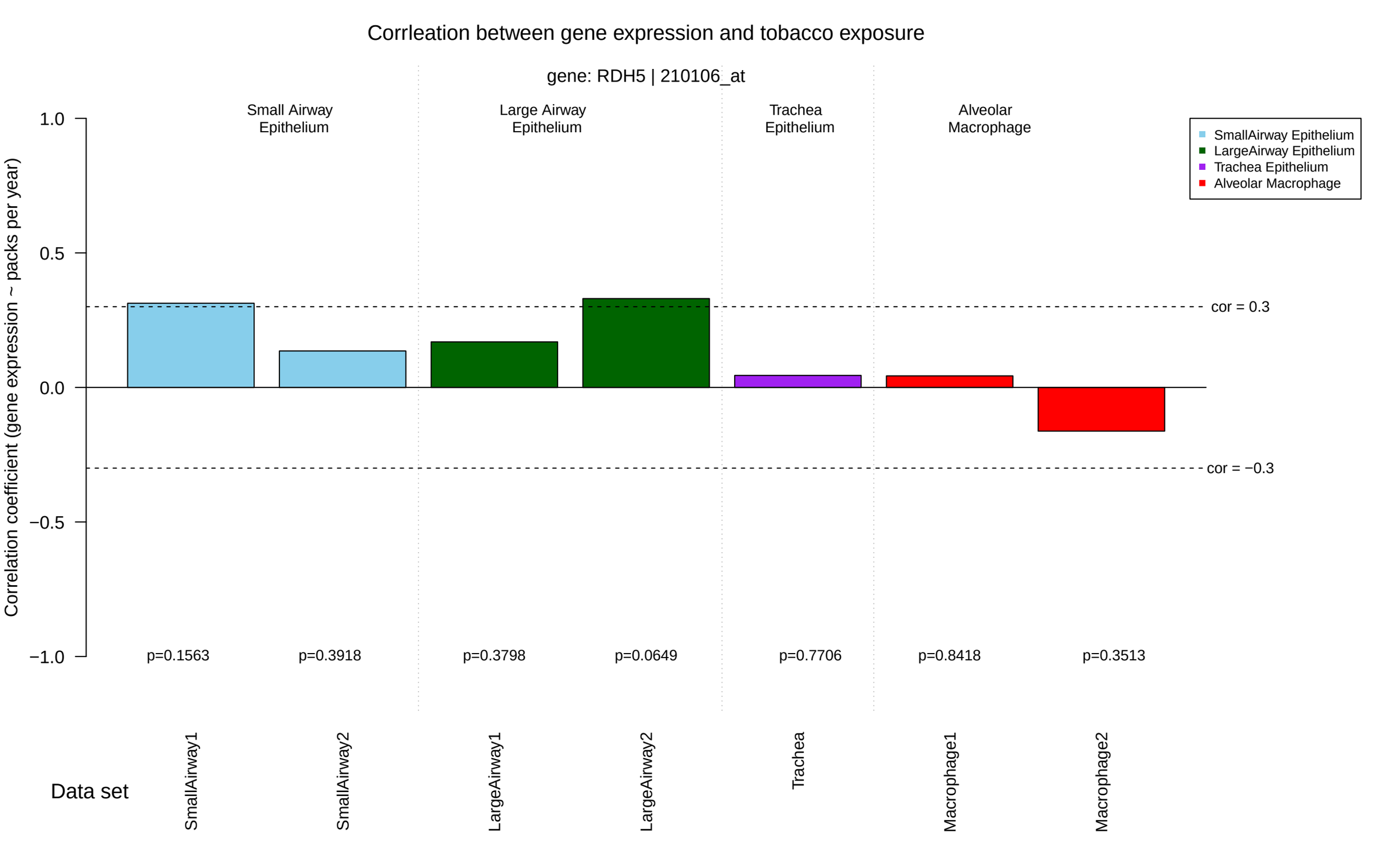

set: LargeAirway1 (p=0.0136)
RDH5 | 210106_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2523)
RDH5 | 210106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6328)
RDH5 | 210106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8765)
RDH5 | 210106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.85)
RDH5 | 210106_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RDH5 | 210106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1563
p=0.3918
p=0.3798
p=0.0649
p=0.7706
p=0.8418
p=0.3513
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2